Telecommunications Churn Prediction and Customer Segmentation
By Jihae Jang

Introduction

Customer churn is an important problem in the telecommunications industry because losing customers can reduce revenue and increase the cost of acquiring replacements. This project aims to predict which customers are likely to churn and to identify meaningful customer segments using the Iranian telecommunications churn dataset, which contains 12 months of customer-level information. The impact of this work is that telecom companies could use churn-risk predictions to prioritize retention efforts and better understand customer behaviour. Rather than treating all customers the same, a company could focus on customers or segments with higher observed churn risk. This project is motivated by the practical value of using machine learning not only to predict customer outcomes, but also to generate business insights that can support decision-making.

I will use both supervised learning and unsupervised learning. For supervised learning, I compare Logistic Regression, Random Forest, and XGBoost. Logistic Regression provides an interpretable baseline, Random Forest captures nonlinear relationships, and XGBoost offers a strongly boosted tree-based model for structured tabular data. The models are evaluated using F1 score, precision, recall, ROC-AUC, and PR-AUC because churn prediction is a binary classification problem where performance on the churn class is especially important. For unsupervised learning, I use clustering to identify hidden customer segments after removing the churn label. K-means is used as an interpretable centroid-based method, while Gaussian Mixture Models provide a probabilistic alternative that can capture softer customer groupings. The clusters are evaluated using silhouette score, cluster size comparison, PCA visualization, and cluster profile analysis, then compared by observed churn rate.

Compared with related telecom churn studies that focus mainly on supervised prediction, this project combines churn prediction with customer segmentation. The main supervised finding is that tree-based ensemble models, especially XGBoost, perform better than simpler baseline models. The main unsupervised finding is that customers can be grouped into distinct behavioural segments with different churn rates. Together, these results show how prediction and segmentation can complement each other in supporting customer retention strategy.

Related Work

A large body of prior work has examined customer churn prediction in the telecommunications industry using machine learning. Wagh et al. (2023) studied telecom churn prediction using machine learning classifiers such as Random Forest, K-Nearest Neighbors, and Decision Tree models[1]. Their work is similar to this project because it focuses on predicting telecom customer churn using supervised classification. However, my project differs by using the Iranian telecommunications churn dataset and by combining supervised churn prediction with unsupervised customer segmentation.

Alotaibi et al. (2024) also examined customer churn prediction in the telecommunications sector using multiple supervised learning models, including Random Forest, LightGBM, XGBoost, Logistic Regression, Decision Trees, and an artificial neural network.[2] Their study is similar to mine because it compares several machine learning models and uses cross-validation and hyperparameter tuning to improve churn prediction. My project is different because it places additional emphasis on model interpretation through feature importance, ablation analysis, threshold sensitivity analysis, and failure analysis.

Wu et al. (2021) proposed an integrated framework that combines churn prediction and customer segmentation for telecommunications customers[3]. This work is highly relevant because it recognizes that churn analysis can benefit from both prediction and segmentation, rather than treating churn only as a classification problem. My project

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[1] Wagh, S. K., Andhale, A. A., Wagh, K. S., Pansare, J. R., Ambadekar, S. P., & Gawande, S. H. (2023). *Customer churn prediction in telecom sector using machine learning techniques*. Results in Control and Optimization, 14, 100342. https://doi.org/10.1016/j.rico.2023.100342
[2] Alotaibi, M. Z., & Haq, M. A. (2024). Customer Churn Prediction for Telecommunication Companies using Machine Learning and Ensemble Methods. *Engineering, Technology & Applied Science Research*, *14*(3), 14572–14578. https://doi.org/10.48084/etasr.7480
[3] Wu, Shuli & Yau, Wei-Chuen & Ong, Thian-Song & Chin, Chong. (2021). Integrated Churn Prediction and Customer Segmentation Framework for Telco Business. IEEE Access. PP. 1-1. 10.1109/ACCESS.2021.3073776

follows a similar combined direction, but it is different as it applies to the Iranian churn dataset and uses XGBoost for supervised prediction along with K-means and Gaussian Mixture Models for unsupervised segmentation.

Data Source

This project uses the Iranian Churn Dataset from the UCI Machine Learning Repository.[4] It contains customer-level telecommunications data collected over a 12-month period from Iranian telecom companies. No external dataset or API was used for this project. The dataset was used as a structured tabular file and loaded into Python for analysis using pandas. The final dataset contains 3,150 customer records and 14 columns. There were no missing values in the loaded dataset, so no imputation was required before modeling. Most variables are integer-valued, with one continuous float variable, Customer Value.

The target variable is *Churn*, which indicates whether a customer left the company or stayed. The main input variables include call behaviour, account characteristics, customer activity, and customer value measures. Important variables include *Call Failure, Complains, Subscription Length, Charge Amount, Seconds of Use, Frequency of Use, Frequency of SMS, Distinct Called Numbers, Age, Status*, and *Customer Value*. These variables were used to support both supervised churn prediction and unsupervised customer segmentation. For supervised learning, the Churn variable was used as the binary outcome to train classification models. For unsupervised learning, the Churn label was removed before clustering so that customer segments were formed based only on input features. After clustering, Churn was added back only for interpretation, allowing the observed churn rate to be compared across customer segments.

Feature Engineering

Before modeling, the dataset was reviewed for missing values, duplicate records, inconsistent data types, and unusual values. Any missing or invalid values were checked to determine whether they should be removed, imputed, or transformed. For example, numeric columns such as age, seconds of use, frequency of use, and charge amount were kept in numeric format, while categorical or binary fields were reviewed to ensure consistent coding. Duplicate rows, if present, were removed to avoid overrepresenting the same customer in the model. Basic exploratory analysis was also used to identify highly redundant variables and features that could create unnecessary noise. This step was important because both supervised and unsupervised models can be affected by irrelevant or duplicate information.

Two variables were removed from the final modeling dataset: *Age Group* and *Tariff Plan*. *Age Group* was dropped because it is redundant with the original *Age* variable. Since age is already available as a continuous numeric feature, including Age Group would duplicate similar information. In addition, tree-based models such as Random Forest and XGBoost can learn age-based thresholds directly from the original Age variable, making the grouped version unnecessary. *Tariff Plan* was also removed because it did not provide meaningful predictive or business value for the final model. Since the goal of the project is to identify customer churn patterns using interpretable behavioral and service-related indicators, variables with limited contribution were excluded to keep the model more focused and easier to explain.

The *Complains* variable indicates whether a customer has previously made a complaint. Because this is a binary categorical feature, it was encoded as: 0 = no complaint and 1 = complaint. One-hot encoding was not required because a single binary column already captures the full information in the variable. Creating separate dummy variables for complaint and no complaint would add redundancy without improving the model. The *Charge Amount* variable was treated as an ordinal feature. It ranges from 0 to 9, where 0 represents the lowest charge amount and 9 represents the highest charge amount. Since the values have a natural order, the feature was kept as an integer rather than one-hot encoded. This allows the model to preserve the ranking information contained in the variable.

A new feature, *Average Call Duration (Seconds)*, was created using the following formula:

[4] UCI Machine Learning Repository. (2020). *Iranian churn dataset*. University of California, Irvine. https://archive.ics.uci.edu/dataset/563/iranian+churn+dataset

Average Call Duration = Seconds of Use / Frequency of Use

This feature represents the average duration of each customer's call. It provides a more interpretable customer usage metric than total seconds of use alone because it captures call behavior at the per-call level. For example, two customers may have the same total usage, but one may make many short calls while another makes fewer long calls. This engineered feature helps the model capture that difference. To avoid division errors, customers with zero frequency of use were checked before creating this feature. If frequency of use was zero, the resulting undefined or infinite values were replaced with an appropriate value, such as 0, depending on the modeling setup.

For supervised learning, the final feature set was used to predict the binary target variable, *Churn*. The target variable itself was excluded from the input features and used only as the outcome label. The final supervised learning features included customer demographics, usage behavior, service interaction, and billing-related information, including all of the feature-engineered features above. These features were used in models such as Logistic Regression, Random Forest, and XGBoost.

For unsupervised learning, the churn label was removed because clustering methods should identify customer segments without being directly trained on the outcome variable. The unsupervised models used selected behavioral and service-related features to group customers based on similarity. Continuous numeric features were scaled before clustering because methods such as K-Means, Gaussian Mixture Models, DBSCAN, and hierarchical clustering are sensitive to feature magnitude. Scaling ensured that features with larger numeric ranges, such as seconds of use, did not dominate the clustering process. Binary or ordinal variables, such as Complains were not used for modelling but just used to interpret the resulting customer segments.

Final supervised learning features include: Age, Call Failure, Complains, Subscription Length, Charge Amount, Seconds of Use, Frequency of Use, Frequency of SMS, Distinct Called Numbers, Age, Customer Value, Average Call Duration (Seconds). Dropped features are: Age Group, Tariff Plan, and Target variable is Churn.

For the unsupervised learning models, the final feature set was limited to numeric customer usage, billing, and demographic variables. The goal was to group customers based on behavioral similarity rather than using the churn outcome or categorical/binary variables that could distort distance-based clustering. The final unsupervised learning features were: Subscription Length, Charge Amount, Seconds of Use, Frequency of Use, Frequency of SMS, Age, Customer Value, Average Call Duration (Seconds).

Before clustering, these numeric features were scaled to ensure that variables with larger ranges, such as Seconds of Use or Customer Value, did not dominate the clustering results. The engineered feature, Average Call Duration, was included because it provides a more interpretable measure of customer call behavior by showing the average duration per call.

The following variables were excluded from the unsupervised clustering models: Churn, Complains, Tariff Plan, Age Group, Call Failure, Status, Distinct Called Numbers. *Churn* was excluded because it is the target variable in the supervised learning task. Including it in clustering would create data leakage and would cause the model to group customers based on the outcome rather than natural customer behavior. *Complains* was excluded because it is a binary variable. Instead of using it directly in the clustering model, it was reserved for post-clustering profiling to help interpret whether certain customer segments had higher complaint rates. *Tariff Plan* was excluded because it is categorical and did not provide meaningful business or predictive contribution for the final clustering analysis. *Age Group* was excluded because it is redundant with the original Age variable. Since Age was already included as a numeric feature, Age Group would duplicate similar information. *Call Failure, Status,* and *Distinct Called Numbers* were excluded to keep the clustering model focused on the most interpretable customer usage, billing, and demographic variables. These features may still be useful for post-clustering analysis, but they were not included in the final clustering input features to reduce noise and improve interpretability.

Part A. Supervised Learning

The supervised learning task was framed as a binary classification problem, where the goal was to predict whether a customer would churn. The target variable was *Churn*, and the input features included customer service indicators, account characteristics, usage behaviour, demographic information, and customer value measures. Before modeling, redundant variables such as *Age Group* and *Tariff Plan* were removed from the final feature set. *Age Group* was dropped because the original *Age* variable was already included, while *Tariff Plan* was removed because it did not provide meaningful predictive or business value in the final model specification. A new feature, *Average Call Duration*, was engineered by dividing *Seconds of Use by Frequency of Use.* The dataset was split into training and test sets using stratification so that the churn and non-churn class proportions were preserved across both sets. This was important because churn was an imbalanced binary outcome, with non-churn customers representing the majority class. The supervised workflow involved fitting multiple model families, comparing their performance of F1 and PR-AUC, selecting the best-performing model, and then tuning that model (which is XGBosst) using cross-validation.

Three supervised model families were explored: Logistic Regression, Random Forest, and XGBoost. Logistic Regression was used as an interpretable baseline model because it provides a simple linear decision boundary and allows the project to compare more complex models against a transparent benchmark. Random Forest was used as a tree-based ensemble method because it can capture nonlinear relationships and interactions among features without requiring strong assumptions about linearity. XGBoost was used as a boosted tree-based ensemble method because it is well suited to structured tabular data and often performs strongly on classification problems with nonlinear feature relationships. The models were evaluated using accuracy, precision, recall, F1 score, ROC-AUC, and PR-AUC. Accuracy was included as a general measure of overall correctness, but it was not treated as the main metric because the churn class was smaller than the non-churn class. Precision and recall were especially important because they describe different business tradeoffs: precision measures how many predicted churners actually churned, while recall measures how many true churners the model successfully identified. F1 score was used because it balances precision and recall. ROC-AUC and PR-AUC were also included to evaluate ranking performance across thresholds, with PR-AUC receiving particular attention because it is more informative when the positive class is relatively uncommon.

Initial model results showed that XGBoost performed best among the three model families. Logistic Regression achieved relatively high recall but low precision, meaning it identified many churners but also produced many false positives. Random Forest improved precision and overall classification performance, but XGBoost produced the strongest balance of precision, recall, F1 score, ROC-AUC, and PR-AUC. Based on this result, XGBoost was selected for deeper evaluation and hyperparameter tuning. Hyperparameter tuning was performed on XGBoost using 5-fold cross-validation and grid search. The tuning process explored combinations of *n_estimators, max_depth,* and *learning_rate,* using PR-AUC as the main scoring metric. PR-AUC was selected because the project focused on identifying churners in an imbalanced classification setting. The best XGBoost parameter combination was then refit on the training data and evaluated on the held-out test set.

The initial supervised model comparison showed that the tree-based ensemble models outperformed the Logistic Regression baseline. Logistic Regression achieved high recall but low precision, suggesting that it was more likely to over-identify customers as churn risks. Random Forest produced stronger precision and F1 performance, showing the benefit of nonlinear tree-based modeling. XGBoost performed best overall, achieving the strongest balance between identifying churners and limiting false positives.

The held-out test-set results also supported XGBoost as the strongest model. In the initial comparison, XGBoost achieved the highest F1 score and PR-AUC among the three models, indicating that it was the most effective at identifying churners while maintaining strong overall ranking performance. After hyperparameter tuning, the tuned XGBoost model maintained strong performance, although the tuned version did not substantially improve over the initial XGBoost model on every metric. This suggests that the default or initial XGBoost configuration was already well suited to the dataset, and that additional tuning mainly confirmed the model’s robustness rather than producing a large performance gain.

```
Fitting 5 folds for each of 60 candidates, totalling 300 fits
Best Parameters:
{'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 500}
Best CV PR-AUC:
0.927065182897237


Tuned XGBoost Results
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.97      0.98       531
           1       0.85      0.91      0.88        99

    accuracy                           0.96       630
   macro avg       0.92      0.94      0.93       630
weighted avg       0.96      0.96      0.96       630

Confusion Matrix:
[[515  16]
 [  9  90]]
ROC-AUC:
0.9931613688675835
PR-AUC:
0.963994547554266
Precision:
0.8490566037735849
Recall:
0.9090909090909091
F1:
0.8780487804878049
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Feature importance was calculated from the tuned XGBoost model. The results show that *Complains* was the most important predictor by a large margin, with an importance score of 0.325. This suggests that whether a customer had previously made a complaint was highly influential in the model's churn predictions. The next most important features were *Frequency of use* with an importance score of 0.123, *Customer Value* with 0.120, and *Seconds of Use* with 0.089. These variables suggest that customer usage behaviour and customer value also played important roles in predicting churn. Other features, including *Charge Amount, Distinct Called Numbers, Age, Subscription Length, Call Failure*, and *Average Call Duration,* contributed more moderately. *Frequency of SMS* had the lowest feature importance, suggesting that SMS activity contributed relatively little to the final XGBoost predictions compared with complaint history and broader usage behaviour.

Overall, the feature-importance results indicate that churn prediction in this dataset was driven most strongly by customer complaint history, followed by usage intensity and customer value. However, feature importance should be interpreted as predictive association rather than causal evidence. The model can show that complaints are useful for predicting churn, but it does not prove that complaints directly cause churn.

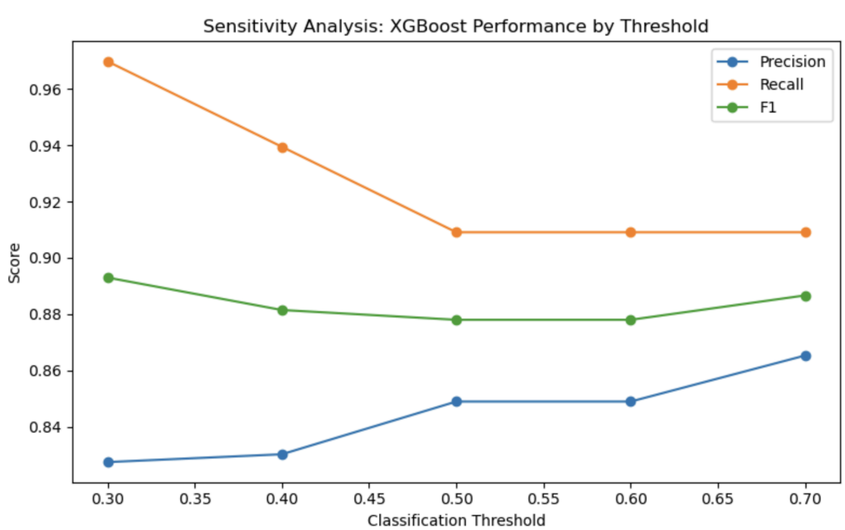
To complement the XGBoost feature-importance results, I conducted an ablation analysis on the tuned XGBoost model. The purpose of this analysis was to test whether removing specific feature groups reduced model performance, which provides stronger evidence about which groups of variables contribute to prediction success. Each ablated model was retrained using the same XGBoost modeling procedure and evaluated on the same held-out test set. The full XGBoost model achieved a precision of 0.849, recall of 0.909, F1 score of 0.878, ROC-AUC of 0.993, and PR-AUC of 0.964. Removing usage features caused the largest performance decline, reducing F1 by 0.133 and PR-AUC by 0.084. This indicates that customer usage behaviour was the most important feature group for predictive performance. These variables included measures such as seconds of use, frequency of use, SMS frequency, distinct called numbers, and average call duration. Removing account and value-related features also reduced performance, lowering F1 by 0.049 and PR-AUC by 0.028. This suggests that subscription length, charge amount, and customer value provided meaningful predictive information. Removing complaint and service-related features reduced F1 by 0.026 and PR-AUC by 0.042. Although Complains had the highest individual feature-importance score, the ablation result shows that the model could still recover some predictive power from other features when complaint information was removed. Removing the demographic feature, Age, had almost no effect on F1, which declined by only 0.001. This suggests that Age was not a major driver of classification performance in the final model. Overall, the ablation analysis shows that usage behaviour contributed most strongly to churn prediction, followed by account/value features and complaint/service indicators. Age contributed the least to prediction success.

Overall, the feature-importance and ablation results show that *Complains* was the most influential individual feature in the XGBoost model. However, the ablation analysis provides a more nuanced interpretation. When complaint/service features were removed, model performance declined, but the decline was smaller than the decline caused by removing the usage-related features. This suggests that, as a group, usage features contributed more to the model's overall predictive performance than complaint information alone. One possible explanation is that complaint behaviour may be indirectly related to usage patterns. For example, customers who experience service issues may change how frequently they use the service, how long they make calls, or how many distinct numbers they contact. As a result, some of the predictive signals captured by the *Complains* variable may also be partially reflected in usage-related variables. Therefore, while complaint history is the strongest single predictor, the broader set of usage features appears to provide the most important feature group for predicting churn.

	Feature Set	Removed Features	Number of Features	Precision	Recall	F1	ROC-AUC	PR-AUC	F1 Change from Full Model	PR-AUC Change from Full Model
0	Full Model	[]	11	0.849057	0.909091	0.878049	0.993161	0.963995	0.000000	0.000000
1	Remove Complaints/Service Features	[Complains, Call Failure]	9	0.786325	0.929293	0.851852	0.985143	0.921863	-0.026197	-0.042131
2	Remove Usage Features	[Seconds of Use, Frequency of use, Frequency o...	6	0.677686	0.828283	0.745455	0.969402	0.880417	-0.132594	-0.083578
3	Remove Account/Value Features	[Subscription Length, Charge Amount, Custome...	8	0.747967	0.929293	0.828829	0.986893	0.935791	-0.049220	-0.028204
4	Remove Demographic Features	[Age]	10	0.823009	0.939394	0.877358	0.988938	0.945118	-0.000690	-0.018877

To assess how sensitive the final XGBoost model was to decision-threshold choice, I evaluated model performance at classification thresholds ranging from 0.30 to 0.70. Although binary classifiers commonly use a default threshold of 0.50, this may not be the best threshold for a churn-retention setting because the business objective may prioritize either identifying more churners or reducing false positives.

The sensitivity analysis showed a clear precision-recall tradeoff. At a lower threshold of 0.30, the model achieved the highest recall of 0.970, meaning it identified the largest share of actual churners. However, precision was lower at 0.828, meaning more non-churners were also flagged as churn risks. At the default threshold of 0.50, precision increased to 0.849, while recall decreased to 0.909. At a higher threshold of 0.70, precision increased further to 0.865, while recall remained at 0.909. These results suggest that the model's overall performance is reasonably stable across thresholds, with F1 scores ranging from 0.878 to 0.893. However, the choice of threshold affects the business tradeoff between broad churn detection and more targeted intervention. If the goal is to identify as many potential churners as possible, a lower threshold such as 0.30 may be preferable. If the goal is to reduce unnecessary retention outreach, a higher threshold such as 0.70 may be more appropriate.



The supervised learning results show several important tradeoffs. The first is the precision-recall tradeoff. In churn prediction, higher recall means the model identifies more customers who actually churn, while higher precision means that customers flagged as churn risks are more likely to truly churn. The threshold sensitivity analysis showed this tradeoff clearly. At a lower threshold of 0.30, recall was highest at 0.970, meaning the model captured more actual churners. However, precision was lower at 0.828, meaning more non-churners were also flagged. At a higher threshold of 0.70, precision increased to 0.865, while recall remained at 0.909. This suggests that the preferred threshold depends on the business objective. If retention outreach is inexpensive, a lower threshold may be useful because it captures more potential churners. If retention interventions are costly, a higher threshold may be better because it focuses on customers with stronger churn signals.

A second tradeoff is between model complexity and interpretability. Logistic Regression is easier to explain because it uses a simpler linear structure, but it did not perform as well as the tree-based models. Random Forest and XGBoost captured more complex nonlinear relationships among customer features and produced stronger performance. However, tree based models such as XGboost and Random Forest are less directly interpretable and require additional tools such as feature importance and ablation analysis to explain which variables contribute to prediction. A third tradeoff is between individual feature importance and feature-group contribution. The XGBoost feature-importance analysis showed that Complains was the strongest individual predictor. However, the ablation analysis showed that removing the full group of usage-related features caused the largest decline in performance. This means that a single feature can appear highly important, while a broader group of related features may contribute more to overall predictive performance. This also suggests that some predictive signals may overlap across variables.

To better understand where the final XGBoost model made mistakes, I reviewed specific test-set records where the predicted churn label did not match the actual churn outcome. This analysis focused on three types of errors: false positives, false negatives, and borderline classification errors. The first type of error was a false positive, where the model predicted that a customer would churn even though the customer actually stayed. These cases may occur when a customer has characteristics that resemble churners, such as complaint history, low usage, or lower customer value, but still remains with the company. In a business setting, false positives may lead to unnecessary retention outreach, but they may be acceptable if the cost of contacting customers is low. The second type of error was a false negative, where the model predicted that a customer would stay even though the customer actually churned. These cases are more concerning for churn prevention because they represent missed opportunities for intervention. False negatives may occur when customers who churn have normal-looking usage patterns or do not show obvious warning signs in the available features. This suggests that the current dataset may be missing important predictors, such as complaint severity, service-resolution history, competitor offers, payment issues, or recent changes in usage behaviour.

The third type of error was a borderline error, where the predicted churn probability was close to the default classification threshold of 0.50. These cases suggest that some customers have ambiguous profiles and are difficult to classify confidently. For these customers, a single hard classification may be less useful than using the predicted probability as a risk score. Overall, the failure analysis shows that the model performs strongly but still struggles with customers whose churn behaviour is not fully explained by the available variables. Future improvements could include adding richer complaint data, complaint-resolution outcomes, customer tenure details, payment history, plan changes, customer service interactions, and time-based changes in usage. These additional features could help distinguish customers who appear similar in the current dataset but have different churn outcomes.

Part B. Unsupervised Learning

The unsupervised learning task is designed to identify hidden customer segments in the Iranian telecommunications churn dataset. Unlike the supervised learning task, the goal is not to predict the churn label directly, but to examine whether customers naturally grouped into distinct behavioural profiles. After clusters were created, the churn label was added back only for interpretation so that observed churn rates could be compared across customer segments. The unsupervised workflow began by creating a separate clustering dataset from the standardized features. The Churn variable was removed before clustering to prevent the algorithms from using the outcome variable when forming customer groups. Variables that were not useful for behavioural segmentation, such as redundant or target-related fields, were excluded where appropriate. The final clustering features focused on customer service, usage, account, demographic, and value-related information. These included variables such as Call Failure, Complains, Subscription Length, Charge Amount, Seconds of Use, Frequency of use, Frequency of SMS, Distinct Called Numbers, Age, Customer Value, Average Call Duration, and call failure rate. Because the clustering features were measured on different scales, all numeric clustering inputs were standardized before modeling. This was necessary because clustering methods such as K-means and Gaussian Mixture Models can be strongly influenced by variables with larger numeric ranges. Standardization helps ensure that variables such as *Seconds of Use* or *Customer Value* do not dominate the clustering process only because of their scale.

Two unsupervised learning methods were explored: K-means clustering and Gaussian Mixture Models (GMM). K-means was selected because it is a simple, widely used centroid-based clustering method that assigns each customer to one hard cluster. This makes it useful for business-facing customer segmentation because each customer belongs to a single, interpretable group. However, K-means assumes that clusters are relatively compact and separated around centroids, which may not fully reflect overlapping customer behaviours. GMM was selected as a probabilistic clustering alternative. Unlike K-means, GMM estimates the probability that each customer belongs to each cluster and can represent softer, overlapping customer segments. This is useful in a churn setting because customers may not fall neatly into one rigid group. For example, a customer may share characteristics with both low-usage and high-value customer groups. Comparing K-means and GMM allowed the project to evaluate whether the customer data was better represented by simple hard segments or by more flexible probabilistic groupings.

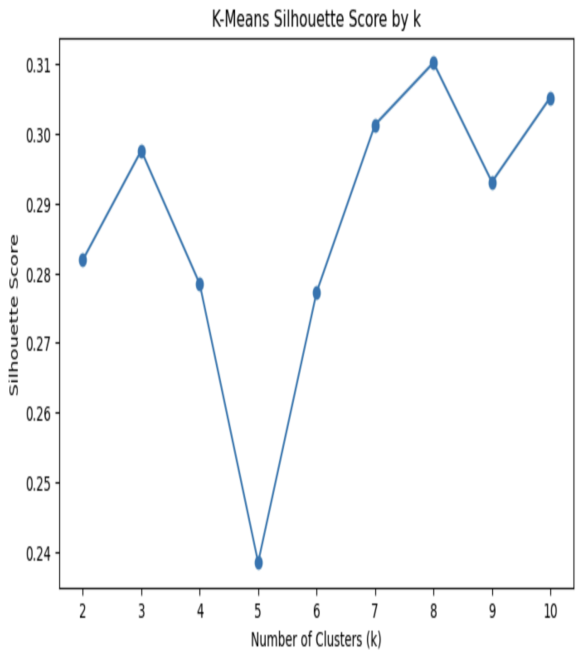
Model exploration was conducted by testing different numbers of clusters or components. For K-means, I evaluated multiple values of k and used metrics such as the elbow method and silhouette score to guide the selection of a reasonable cluster solution. Cluster size and interpretability were also considered because a technically valid cluster solution is less useful if it produces too many small or difficult-to-explain groups. For GMM, I compared different numbers of mixture components and used the Bayesian Information Criterion (BIC) to identify the best-fitting model. In the notebook results, the best GMM solution by BIC used 10 components, which produced customer groups with different observed churn and complaint rates. After fitting the clustering models, I profiled the resulting clusters by calculating average feature values within each cluster. I then compared observed churn rates across clusters. This step did not use churn to create the clusters; instead, churn was used after clustering as an external interpretation variable. This allowed the analysis to identify whether some customer segments were more strongly associated with churn than others. PCA was also used for visualization so that the high-dimensional clustering results could be displayed in two dimensions.

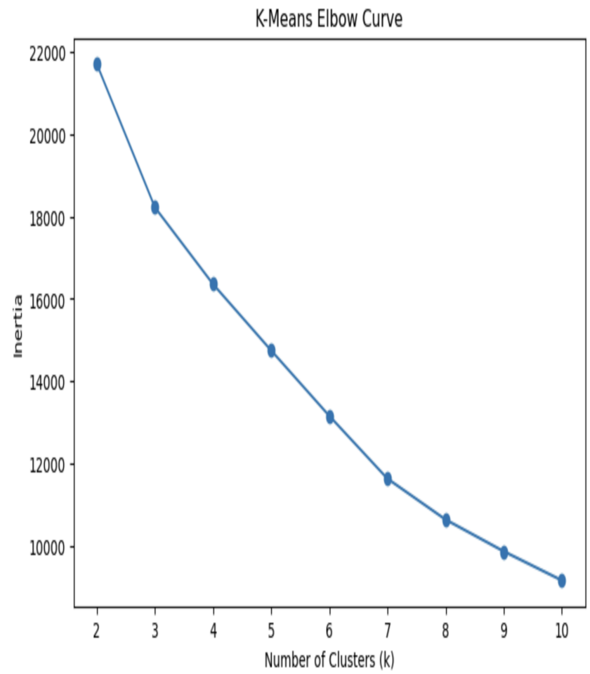
The unsupervised models were evaluated using a combination of quantitative clustering metrics, visual inspection, and business interpretability. Because clustering does not have a true target label, evaluation focused on whether the models created compact, separated, and interpretable customer groups. After clusters were created, the churn label was added back only for interpretation so that observed churn rates could be compared across segments.

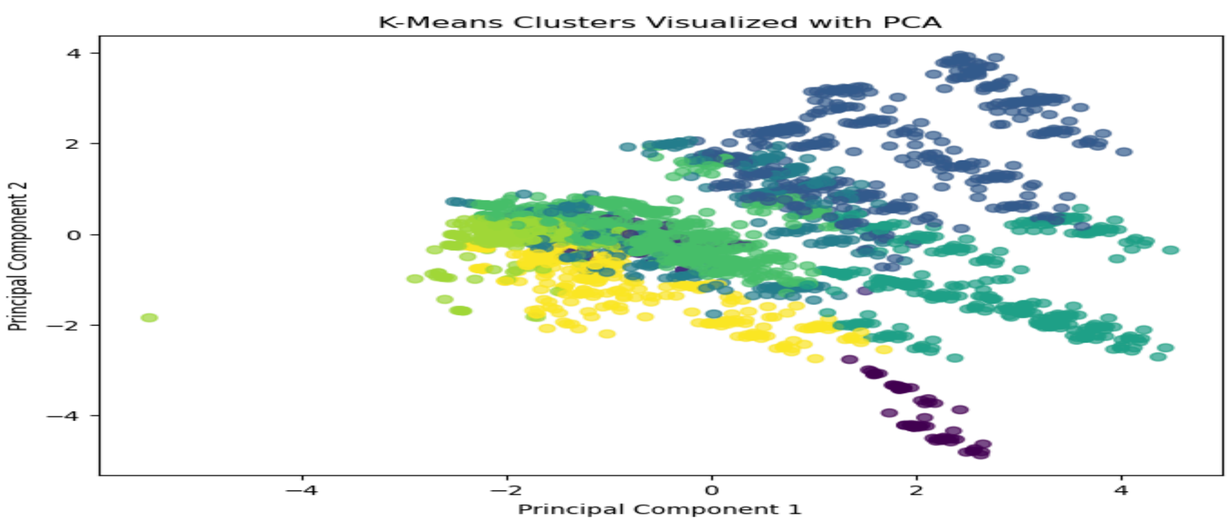
The main quantitative metrics used were silhouette score, Davies-Bouldin index, Calinski-Harabasz score, and, for Gaussian Mixture Models, Bayesian Information Criterion (BIC). The silhouette score measures how well each customer fits within its assigned cluster compared with other clusters, where higher values indicate better separation. The Davies-Bouldin index measures average cluster similarity, where lower values indicate better-defined clusters. The Calinski-Harabasz score compares between-cluster separation with within-cluster compactness, where higher values indicate stronger clustering structure. For GMM, BIC was used to compare models with different numbers of components while penalizing unnecessary model complexity. K-means and GMM were compared because they represent different clustering approaches. K-means is a centroid-based method that assigns each customer to one hard cluster, making it simple and interpretable for customer segmentation. GMM is a probabilistic method that allows customers to have softer membership across overlapping groups. This made GMM useful for testing whether customer behaviour was better represented by more flexible segments.

The overall comparison showed that both methods identified meaningful customer segments, but they differed in complexity and interpretability. K-means produced simpler hard clusters that were easier to explain as customer segments. GMM produced a more flexible segmentation structure and identified groups with different observed churn rates, but the larger number of components made the results more complex to interpret. Therefore, the best model was selected not only based on metric performance, but also based on whether the resulting clusters were understandable and useful for business interpretation.

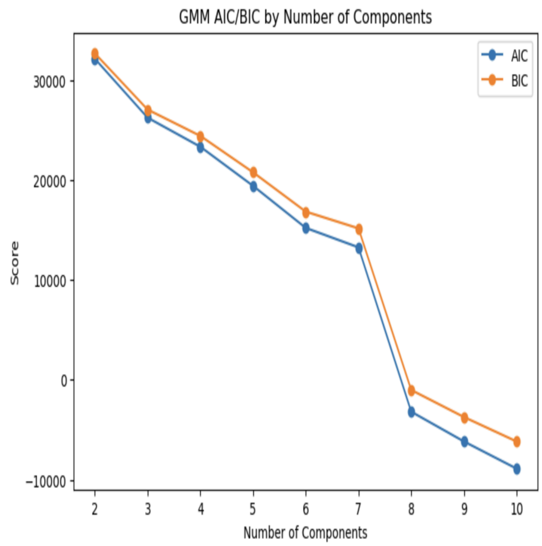
For K-means, the first visualization was the elbow or silhouette plot used to evaluate different values of k. This helped determine whether adding more clusters improved the clustering structure enough to justify additional complexity. The second visualization was a PCA scatterplot showing customers projected into two dimensions and colored by the K-means cluster. This visualization helped assess whether the clusters were visually separated and whether the chosen solution produced interpretable groupings.

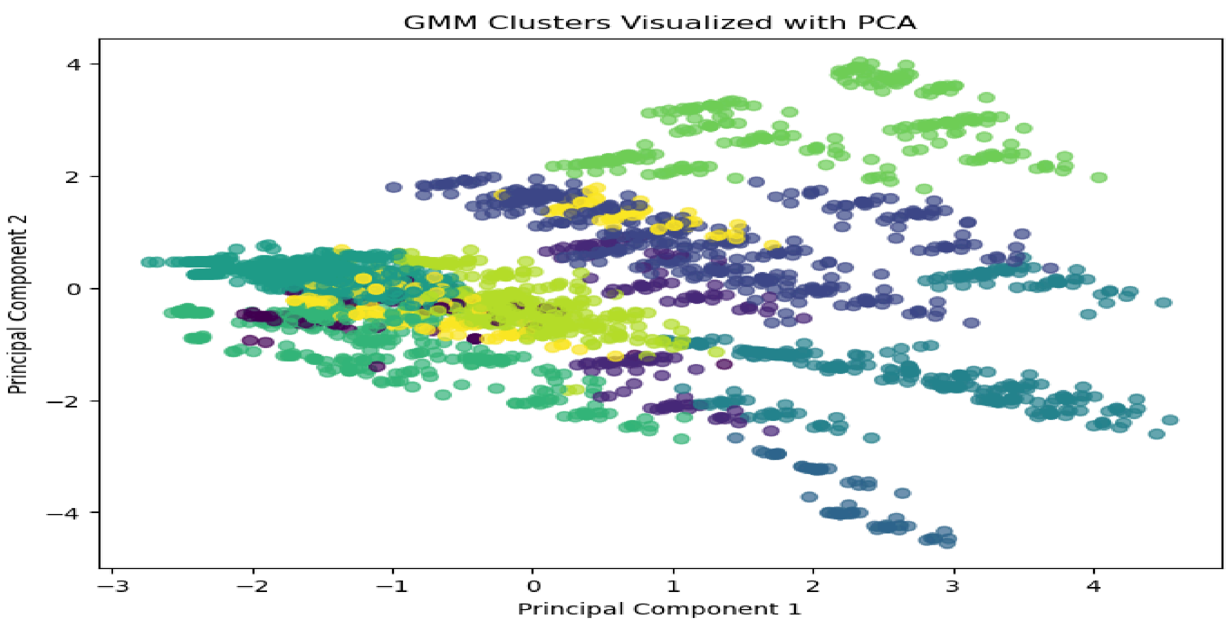






For GMM, the first visualization was the BIC plot across different numbers of mixture components. This plot was used to evaluate model fit while penalizing complexity. In the notebook results, BIC selected a 10-component GMM solution, suggesting that the customer data may contain multiple overlapping behavioural profiles. The second visualization was a PCA scatterplot colored by the GMM cluster, which helped show how the probabilistic clusters appeared in reduced two-dimensional space.





Cluster profile analysis was also used to interpret the customer segments. For each cluster, I compared average values of key features such as complaint rate, usage behaviour, customer value, and observed churn rate. This analysis was important because clustering metrics alone do not explain whether a segmentation solution is meaningful for a business problem. A useful clustering model should create groups that are not only mathematically distinct, but also interpretable in terms of customer behaviour. The cluster profiles showed that observed churn rates differed across customer segments. This suggests that the clustering models captured behaviourally meaningful differences among customers. However, these results should be interpreted as associations rather than causal findings. Since churn was not used to create the clusters, differences in churn rate across clusters suggest that some customer profiles are more strongly associated with churn, but they do not prove that cluster membership causes churn.

To assess the stability of the unsupervised results, I conducted a sensitivity analysis by varying the number of clusters/components. This tested whether the segmentation findings depended heavily on a single selected value of k. For K-means, I compared multiple k values and reviewed changes in silhouette score, cluster size, and interpretability. For GMM, I compared different numbers of mixture components using BIC and examined whether the resulting cluster profiles remained meaningful.

```
Features used for GMM:
['Subscription  Length', 'Charge  Amount', 'Seconds of Use', 'Frequency of use', 'Frequency of SMS', 'Age', 'Customer Value', 'Avg_Call_Dur
ation_Sec']
GMM Evaluation Results:
```

	k	AIC	BIC	Silhouette Score	Davies-Bouldin Score	Calinski-Harabasz Score
0	2	32126.947419	32665.856457	0.270369	1.660770	941.872202
1	3	26230.981724	27042.372860	0.287875	1.607759	824.933682
2	4	23324.235148	24408.108382	0.326286	1.326505	882.147269
3	5	19409.688520	20766.043852	0.337953	1.221766	831.001694
4	6	15199.357647	16828.195077	0.305983	1.455832	750.009893
5	7	13253.853482	15155.173010	0.317094	1.190849	838.994326
6	8	-3197.091152	-1023.289526	0.196567	1.340639	721.125972
7	9	-6175.096809	-3728.813086	0.200821	1.348709	683.042489
8	10	-8895.932168	-6177.166347	0.240349	1.274187	739.721949

The sensitivity analysis showed that increasing the number of clusters generally allowed the models to capture more detailed customer profiles, but it also made the results harder to interpret. A smaller number of clusters produced simpler segments, but sometimes combined customers with meaningfully different usage or churn patterns. A larger number of clusters, especially in GMM, captured more detailed variation in churn and complaint rates, but some segments became smaller and less practical for business interpretation. This reveals an important tradeoff in unsupervised learning: the best statistical clustering solution is not always the most useful business segmentation. For this project, the final segmentation choice balanced clustering quality, churn-rate separation, cluster size, and interpretability.

Discussion

Part A: Supervised Learning

The supervised learning portion of the project showed that customer churn can be predicted effectively using behavioural, service-related, account, and value-based features. One key lesson was that model choice matters for this type of structured tabular data. Logistic Regression provided a useful baseline, but the tree-based ensemble models performed better, especially XGBoost. This suggests that the relationship between customer characteristics and churn is not purely linear and may involve interactions among features such as complaint history, usage behaviour, and customer value. One surprising result was the difference between individual feature importance and feature-group importance. The XGBoost feature-importance results showed that *Complains* was the strongest individual predictor. However, the ablation analysis showed that removing the full group of usage-related features caused the largest decline in model performance. This suggests that complaint history is highly informative on its own, but broader usage behaviour contributes more to the model's overall predictive power. It may also indicate that complaint behaviour and usage patterns contain overlapping predictive signals.

A major challenge in the supervised analysis was choosing the right evaluation metrics. Accuracy alone was not sufficient because churn prediction is an imbalanced binary classification problem, and the business goal is especially focused on identifying customers who are likely to churn. To respond to this challenge, I evaluated the models using precision, recall, F1 score, ROC-AUC, and PR-AUC. I also conducted threshold sensitivity analysis to examine how changing the classification threshold affected the balance between precision and recall. With more time and resources, I would extend the supervised model by adding richer customer-service and time-based features. For example, the current *Complains* variable only indicates whether a complaint occurred, but it does not capture complaint type, severity, timing, resolution status, or number of follow-ups. I would also add monthly usage trends instead of relying only on aggregate usage measures. These additions could help distinguish customers who look similar in the current dataset but have different churn outcomes.

Part B: Unsupervised Learning

The unsupervised learning portion showed that clustering can complement supervised prediction by identifying broader customer segments. While supervised learning estimates individual churn risk, unsupervised learning helps describe different types of customers and compare their observed churn rates. This was useful because it moved the project beyond prediction alone and provided a more interpretable customer-segmentation perspective. One important lesson was that clustering results must be evaluated using both technical metrics and business interpretability. K-means produced simpler hard clusters that were easier to explain, while Gaussian Mixture Models provided a more flexible probabilistic segmentation. The GMM results suggested that customers may not fall into only a few simple groups; instead, there may be several overlapping behavioural profiles with different churn and complaint patterns. A surprising result was that the statistically preferred GMM solution used a relatively large number of components. This suggested that the customer base may contain more detailed behavioural variation than a simple low-, medium-, and high-risk segmentation would capture. However, this also created an interpretation challenge. More clusters can capture more detail, but too many clusters may be harder to explain or use in a business setting.

The main challenge in the unsupervised analysis was selecting a cluster solution that was both technically reasonable and practically meaningful. Metrics such as silhouette score, Davies-Bouldin index, Calinski-Harabasz score, and BIC were useful, but they did not fully answer whether the segments were useful for retention strategy. To respond to this, I also examined cluster sizes, PCA visualizations, cluster profiles, and observed churn rates by cluster. With more time and resources, I would extend the unsupervised analysis by testing additional clustering methods and richer feature representations. I would also validate whether the customer segments remain stable across different time periods or new customer samples. Finally, I would connect the clusters to specific retention strategies, such as complaint-resolution outreach, usage-based engagement campaigns, or high-value customer retention programs.

Ethical Considerations

Part A: Supervised Learning

Several ethical issues could arise from using a supervised churn prediction model. First, there is a risk that the model could be used to make unfair or overly aggressive decisions about customers. For example, customers predicted to be likely churners could receive different treatment, pricing, or service attention than customers predicted to stay. If the model relies on variables that are correlated with demographic or socioeconomic characteristics, such as age, usage level, or customer value, it could unintentionally reinforce unequal treatment across customer groups.

A second issue is the risk of treating predictive associations as causal explanations. For example, the model may find that complaint history is strongly associated with churn, but that does not prove that complaints directly cause churn. Customers may complain because of service issues, billing problems, or other factors not captured in the dataset. To address this, the results should be framed as predictive rather than causal, and any retention intervention should be tested carefully before being deployed. Finally, false positives and false negatives have different ethical and business consequences. A false positive may lead to unnecessary outreach, while a false negative may cause a company to miss a customer who needed support. To reduce harm, the model should be used as a decision-support tool rather than an automatic decision-maker. Human review, transparent communication, and ongoing monitoring would be important before applying this type of model in a real customer-service setting.

Part B: Unsupervised Learning

The main ethical issue in the unsupervised learning component is the risk of labeling or profiling customer groups too simplistically. Clustering assigns customers into segments based on behavioural and account-related features, but those segments do not fully represent individual customers. If a company treats cluster membership as a fixed customer identity, it could lead to stereotyping or inappropriate business actions.

Another issue is that clusters may reflect patterns in the dataset that are not stable or meaningful outside the sample. For example, a cluster with a high observed churn rate may appear important, but that does not mean every customer in that cluster is likely to churn or that the cluster would remain the same in future data. To address this, cluster results should be validated on additional data and interpreted as exploratory customer profiles rather than definitive customer categories. There is also a privacy concern. Even though the dataset used in this project is public and does not include direct identifiers, real-world customer segmentation often involves sensitive behavioural data. Usage patterns, complaints, payment behaviour, and service interactions can reveal personal information about customers. A real deployment would require data minimization, secure storage, and clear limits on how customer data is used. To address these concerns, the clustering results should be used carefully and transparently. Segments should support broad service improvement and retention planning, not punitive or discriminatory treatment. The analysis should also avoid causal claims, since clustering only identifies patterns of similarity and association. Overall, both the supervised and unsupervised components should be treated as tools for supporting better customer understanding, not as automated systems for making final decisions about individual customers.

Statement of Work

This project was completed independently by Jihae Jang. I was responsible for all parts of the project, including selecting the dataset, conducting exploratory data analysis, cleaning and preparing the data, engineering features, building and evaluating the supervised learning models, conducting the unsupervised clustering analysis, interpreting the results, creating visualizations, writing the final report, and preparing the GitHub repository and project submission materials.

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